

# **ESAT CAMP MOCK TEST C**

Physics

EC-PHY-C

**PHYSICS**

**40 MINUTES**

## **INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## Physics

- 1 A magnet rests on a top-pan balance. A straight horizontal wire of length  $4.0\text{ cm}$  is held between the magnet's poles and carries current  $I$ . The balance reading increases linearly from  $500.0\text{ g}$  at  $I = 0$  to  $504.0\text{ g}$  at  $I = 2.0\text{ A}$ . What is the magnitude of the magnetic field strength between the poles? Take  $g = 10\text{ N kg}^{-1}$ .
- A 0.0050 T
  - B 0.050 T
  - C 0.50 T
  - D 2.0 T
  - E 5.0 T
  - F 50 T

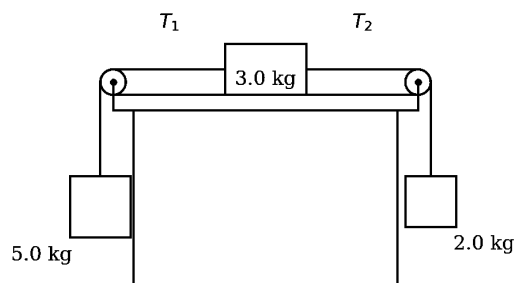
- 2 A block of mass 3.0 kg rests on a smooth, horizontal table. It is connected by two light, inextensible strings passing over smooth, frictionless pulleys at opposite ends of the table to two hanging masses of 5.0 kg and 2.0 kg, as shown in the diagram.

The system is released from rest.

What is the tension  $T_1$  in the string attached to the 5.0 kg mass and the tension  $T_2$  in the string attached to the 2.0 kg mass while the blocks are moving?

[Take the acceleration due to gravity to be  $g = 10 \text{ m s}^{-2}$ .]

|   | $T_1 / \text{N}$ | $T_2 / \text{N}$ |
|---|------------------|------------------|
| A | 15               | 6                |
| B | 15               | 20               |
| C | 35               | 20               |
| D | 35               | 26               |
| E | 50               | 20               |
| F | 50               | 26               |
| G | 65               | 14               |
| H | 65               | 20               |



*Diagram not to scale*

- 3 A laser beam reflects off a plane mirror  $M_1$  and then off a second plane mirror  $M_2$ . The direction of the beam incident on  $M_1$  is fixed.  $M_1$  is rotated by an angle  $\alpha$  in a clockwise direction.

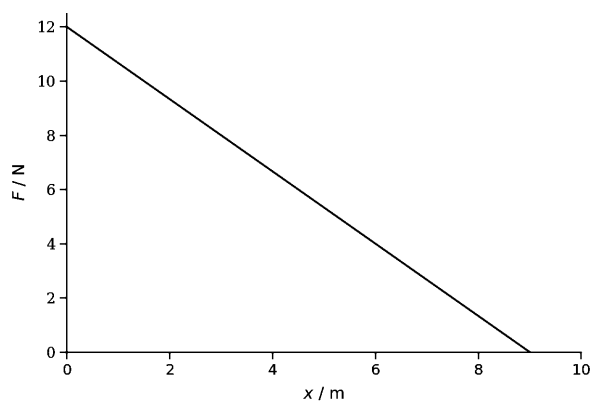
To ensure the final beam reflected from  $M_2$  maintains a constant direction,  $M_2$  must be rotated by:

- A  $\alpha$  in the clockwise direction
- B  $\alpha$  in the anticlockwise direction
- C  $2\alpha$  in the clockwise direction
- D  $2\alpha$  in the anticlockwise direction
- E  $\frac{\alpha}{2}$  in the clockwise direction

- 4 A block of mass  $2.0 \text{ kg}$  is initially moving at a speed of  $4.0 \text{ m s}^{-1}$  along a smooth horizontal surface in the positive  $x$ -direction.

A single horizontal force  $F$  acts on the block in the direction of its motion. The graph shows how  $F$  varies with the displacement  $x$  of the block.

What is the speed of the block when it reaches  $x = 6.0 \text{ m}$ ?



- A  $6.0 \text{ m s}^{-1}$
- B  $6.9 \text{ m s}^{-1}$
- C  $7.2 \text{ m s}^{-1}$
- D  $8.0 \text{ m s}^{-1}$
- E  $8.4 \text{ m s}^{-1}$
- F  $9.4 \text{ m s}^{-1}$

- 5 A spring obeys Hooke's law. A force of 6.0 N produces an extension of 3.0 cm.

The spring is already extended by 2.0 cm. How much additional energy must be transferred to the spring to increase its extension to 5.0 cm?

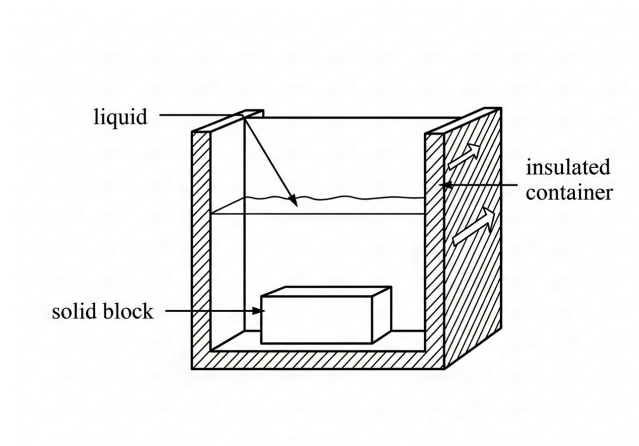
- A 0.09 J
- B 0.15 J
- C 0.21 J
- D 0.25 J
- E 0.29 J

- 6 A hot solid block is placed into a cold liquid in an insulated container.

The mass of the liquid is twice the mass of the block. The specific heat capacity of the liquid is three times the specific heat capacity of the block.

The temperature of the block decreases by  $\Delta T$ .

Assuming no thermal energy is transferred to the surroundings, what is the temperature increase of the liquid?



- A  $\frac{\Delta T}{6}$
- B  $\frac{\Delta T}{5}$
- C  $\frac{\Delta T}{3}$
- D  $\frac{\Delta T}{2}$
- E  $5\Delta T$
- F  $6\Delta T$

- 7 The activity of a radioactive source is measured. Its activity after 1 hour is 8 times its activity after 4 hours.

Its activity after 2 hours is 120 Bq.

What was the initial activity of the source in Bq?

- A 30
- B 120
- C 240
- D 480
- E 960
- F 1920

- 8 A sealed syringe with a freely moving, frictionless piston contains a fixed mass of an ideal gas.

At a depth of 30 m below the surface of a lake, the volume of the gas in the syringe is  $80 \text{ cm}^3$ .

When the syringe is moved to a depth of 10 m below the surface, the volume of the gas increases to  $120 \text{ cm}^3$ .

The temperature of the water is uniform throughout the lake, and the gas remains in thermal equilibrium with the water.

What is the volume of the gas in the syringe at the surface of the lake?

- A  $135 \text{ cm}^3$
- B  $140 \text{ cm}^3$
- C  $150 \text{ cm}^3$
- D  $160 \text{ cm}^3$
- E  $180 \text{ cm}^3$
- F  $200 \text{ cm}^3$
- G  $240 \text{ cm}^3$
- H  $360 \text{ cm}^3$

- 9** A detector records the mean count rates shown for a source emitting two types of radiation. Its background count rate is  $20 \text{ counts min}^{-1}$ .

The paper stops all alpha particles. The aluminium stops all alpha and beta particles. Neither absorber significantly affects gamma radiation.

Which two types are detected from the source, and what is the count rate due to the more penetrating type alone?

| Absorber  | Mean count rate / $\text{counts min}^{-1}$ |
|-----------|--------------------------------------------|
| none      | 240                                        |
| paper     | 240                                        |
| aluminium | 90                                         |

- A** alpha and beta;  $70 \text{ counts min}^{-1}$   
**B** alpha and gamma;  $90 \text{ counts min}^{-1}$   
**C** beta and gamma;  $70 \text{ counts min}^{-1}$   
**D** beta and gamma;  $90 \text{ counts min}^{-1}$   
**E** beta and gamma;  $150 \text{ counts min}^{-1}$
- 10** A  $0.050 \text{ kg}$  block is launched along a horizontal track by a spring of spring constant  $800 \text{ N m}^{-1}$ , initially compressed by  $5.0 \text{ cm}$ . The spring obeys Hooke's law and is not attached to the block.

The block loses  $0.20 \text{ J}$  of energy on a rough section, then rises up a smooth slope. All other energy losses are negligible.

What maximum vertical height does it reach above its starting level? Take  $g = 10 \text{ N kg}^{-1}$ .

- A**  $0.40 \text{ m}$   
**B**  $0.80 \text{ m}$   
**C**  $1.0 \text{ m}$   
**D**  $1.6 \text{ m}$   
**E**  $2.0 \text{ m}$

- 11** A U-tube of uniform cross-sectional area, open to the atmosphere at both ends, initially contains a liquid of density  $\rho_1$ .

A second immiscible liquid of density  $\rho_2$  is poured into the left arm, forming a column of height  $h$ .

A third immiscible liquid of density  $\rho_3$  is then poured into the right arm until the top liquid surfaces in both arms are at the exact same horizontal level, as shown in the diagram.

Given that  $\rho_1 > \rho_2 > \rho_3$ , which expression gives the height  $h_3$  of the column of the third liquid?

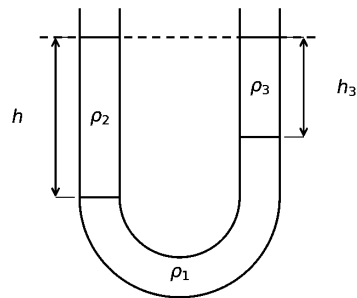


Diagram not to scale

- A**  $\left(\frac{\rho_2}{\rho_3}\right) h$
- B**  $\left(\frac{\rho_3}{\rho_2}\right) h$
- C**  $\left(\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}\right) h$
- D**  $\left(\frac{\rho_1 - \rho_3}{\rho_1 - \rho_2}\right) h$
- E**  $\left(\frac{\rho_2 - \rho_3}{\rho_1 - \rho_3}\right) h$
- F**  $\left(\frac{\rho_1 - \rho_2}{\rho_2 - \rho_3}\right) h$
- G**  $\left(\frac{\rho_1 + \rho_2}{\rho_1 + \rho_3}\right) h$
- H**  $\left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_3}\right) h$

- 12** Plane water waves travel along a straight canal. At a stationary post, wave crests are observed to pass every 2.0 s. A boat travels along the canal in the same direction as the waves with a constant speed of 4.0 m/s. An observer on the boat records that wave crests overtake the boat every 6.0 s.

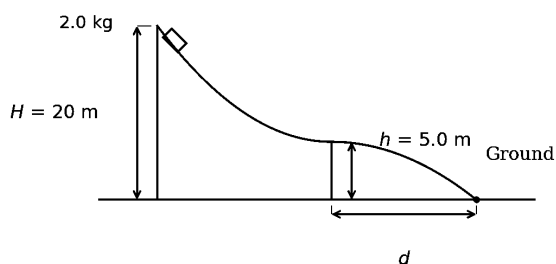
What is the speed of the waves?

- A** 2.0 m/s  
**B** 3.0 m/s  
**C** 4.0 m/s  
**D** 6.0 m/s  
**E** 12.0 m/s
- 13** A small block of mass 2.0 kg is released from rest at the top of a smooth, curved ramp at a height  $H = 20$  m above horizontal ground. The end of the ramp is horizontal and is at a height  $h = 5.0$  m above the ground, where the block leaves the ramp.

Air resistance is negligible.

The gravitational field strength is  $g = 10 \text{ N kg}^{-1}$ .

What is the horizontal distance  $d$  travelled by the block between leaving the ramp and landing on the ground?



*Diagram not to scale*

- A**  $5\sqrt{3}$  m  
**B** 10 m  
**C**  $10\sqrt{3}$  m  
**D** 15 m  
**E** 20 m  
**F**  $20\sqrt{3}$  m

- 14** Three identical resistors are connected in a circuit with a DC power supply. One resistor is connected in series with a parallel combination of the other two.  
What fraction of the total power supplied by the source is dissipated by the parallel combination?
- A**  $\frac{1}{9}$   
**B**  $\frac{1}{6}$   
**C**  $\frac{1}{4}$   
**D**  $\frac{1}{3}$   
**E**  $\frac{1}{2}$   
**F**  $\frac{2}{3}$
- 15** An electric pump lifts oil of density  $800 \text{ kg m}^{-3}$  through a vertical height of  $5.0 \text{ m}$  at a constant volumetric flow rate of  $2.5 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$ . The pump system has an overall efficiency of 50% and is connected to a  $50 \text{ V}$  DC power supply.  
Assuming  $g = 10 \text{ m s}^{-2}$ , what is the current drawn from the supply?
- A**  $1.0 \text{ A}$   
**B**  $2.0 \text{ A}$   
**C**  $2.5 \text{ A}$   
**D**  $4.0 \text{ A}$   
**E**  $5.0 \text{ A}$
- 16** A continuous wave travels from medium P, where its speed is  $v$ , into medium Q, where its speed is  $0.8v$ .  
Which of the following gives the fractional change in the wavelength of the wave?
- A**  $-\frac{1}{4}$   
**B**  $-\frac{1}{5}$   
**C**  $0$   
**D**  $\frac{1}{5}$   
**E**  $\frac{1}{4}$

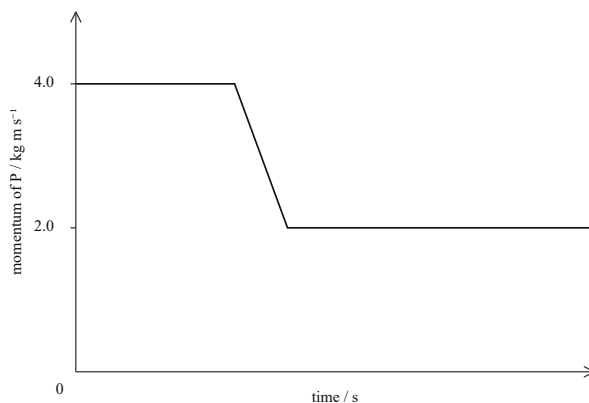
- 17** A solid cuboid is made from a material of density  $2000 \text{ kg m}^{-3}$ . Its side lengths are in the ratio 1 : 2 : 4. The cuboid is placed on a horizontal surface to exert the maximum possible pressure. This pressure is 8.0 kPa.

What is the mass of the cuboid?

(The gravitational field strength  $g$  is  $10 \text{ N kg}^{-1}$ )

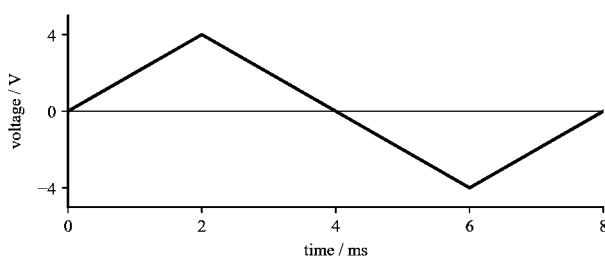
- A** 2.0 kg
  - B** 16 kg
  - C** 128 kg
  - D** 160 kg
  - E** 1024 kg
  - F** 16000 kg
- 18** An NTC thermistor is connected in series with a  $2.0 \text{ k}\Omega$  resistor across a constant 12 V supply. At the initial temperature the thermistor resistance is  $4.0 \text{ k}\Omega$ .
- The thermistor is warmed until its resistance becomes  $1.0 \text{ k}\Omega$ . What happens to the electrical power transferred to the thermistor?
- A** It decreases to one quarter of its initial value.
  - B** It decreases to one half of its initial value.
  - C** It remains unchanged.
  - D** It doubles.
  - E** It quadruples.

- 19** An object P collides with a stationary object Q on a straight, frictionless track. The graph shows the momentum of object P as a function of time.



What is the momentum of object Q after the collision?

- A**  $-4.0 \text{ kg m s}^{-1}$
  - B**  $-2.0 \text{ kg m s}^{-1}$
  - C**  $0 \text{ kg m s}^{-1}$
  - D**  $2.0 \text{ kg m s}^{-1}$
  - E**  $4.0 \text{ kg m s}^{-1}$
  - F**  $6.0 \text{ kg m s}^{-1}$
- 20** The graph shows the voltage produced by an electromagnetic generator during one complete cycle. It is connected in series with a  $2.0 \text{ k}\Omega$  resistor and an ideal diode. The diode allows current only when the plotted voltage is positive. The voltage is unchanged by the connection.
- How much charge passes through the resistor during one complete cycle?



- A**  $2.0 \mu\text{C}$
- B**  $4.0 \mu\text{C}$
- C**  $8.0 \mu\text{C}$
- D**  $16 \mu\text{C}$
- E** zero

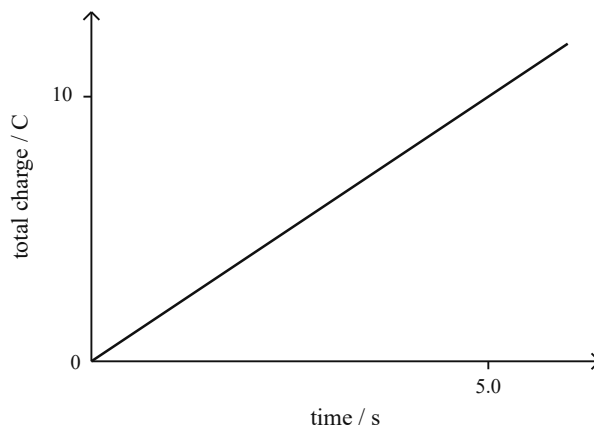
- 21** An LDR and a fixed  $1.0 \text{ k}\Omega$  resistor are connected in series across an ideal  $9.0 \text{ V}$  supply. At the fixed light intensity, the working LDR has resistance  $2.0 \text{ k}\Omega$ .
- A fault develops in exactly one component. An ideal voltmeter across the LDR now reads  $9.0 \text{ V}$ , and an ideal ammeter in series with the supply reads  $4.5 \text{ mA}$ . All connecting wires remain intact. Which fault is consistent with both readings?
- A** The LDR has become an open circuit.
  - B** The LDR has become a short circuit.
  - C** The fixed resistor has become an open circuit.
  - D** The fixed resistor has become a short circuit.
  - E** The LDR resistance has increased to  $3.0 \text{ k}\Omega$ .
- 22** An ideal gas bubble of volume  $V$  is released at a depth  $3H$  below the surface of a liquid. It rises to a depth  $H$ .
- The atmospheric pressure at the surface is equal to the pressure exerted by a column of the liquid of height  $H$ .
- Assuming the temperature of the gas remains constant, what is the volume of the bubble at depth  $H$ ?
- A**  $\frac{V}{3}$
  - B**  $\frac{V}{2}$
  - C**  $2V$
  - D**  $3V$
  - E**  $4V$
- 23** A laser beam from a fixed source is incident on a plane mirror  $M_1$ . The reflected beam then strikes a second plane mirror  $M_2$ .  $M_1$  rotates with angular speed  $\omega$  and  $M_2$  rotates with angular speed  $4\omega$ . Both mirrors rotate in the same direction about parallel axes perpendicular to the plane of incidence.
- What is the angular speed of the final beam reflected from  $M_2$ ?
- A**  $3\omega$
  - B**  $5\omega$
  - C**  $6\omega$
  - D**  $8\omega$
  - E**  $10\omega$

- 24** A solid cuboid of mass  $1.0 \text{ kg}$  is placed on a horizontal surface. Depending on which face it rests on, it can exert one of three possible pressures:  $0.40 \text{ N cm}^{-2}$ ,  $0.50 \text{ N cm}^{-2}$ , or  $2.0 \text{ N cm}^{-2}$ .

What is the density of the cuboid's material, in  $\text{g cm}^{-3}$ ?

(Take the gravitational field strength  $g$  to be  $10 \text{ N kg}^{-1}$ .)

- A** 0.05  
**B** 0.40  
**C** 2.0  
**D** 2.5  
**E** 20  
**F** 20000
- 25** A resistor of resistance  $4.0 \Omega$  is connected to a power supply. The graph shows the total charge that has passed through the resistor as a function of time.



What is the potential difference across the resistor?

- A** 0.50 V  
**B** 2.0 V  
**C** 8.0 V  
**D** 20 V  
**E** 25 V  
**F** 40 V

- 26** A charged particle of mass  $m$  is released from rest in a uniform field. The particle experiences a constant force  $F$  due to the field.

What is the kinetic energy of the particle after it has travelled a distance  $d$  parallel to the force?

- A**  $\frac{1}{2}Fd$
- B**  $2Fd$
- C**  $Fd$
- D**  $\frac{Fd}{m}$
- E**  $Fdm$

- 27** An insulated metal tank acquires a positive charge as fuel flows out. It is then connected to Earth by a conducting wire and becomes uncharged.

Which row correctly describes the charging process and the particle movement through the wire during discharge?

- A** The tank gained protons; protons move from the tank to Earth.
- B** The tank lost electrons; electrons move from the tank to Earth.
- C** The tank gained electrons; electrons move from Earth to the tank.
- D** The tank lost protons; protons move from Earth to the tank.
- E** The tank lost electrons; electrons move from Earth to the tank.